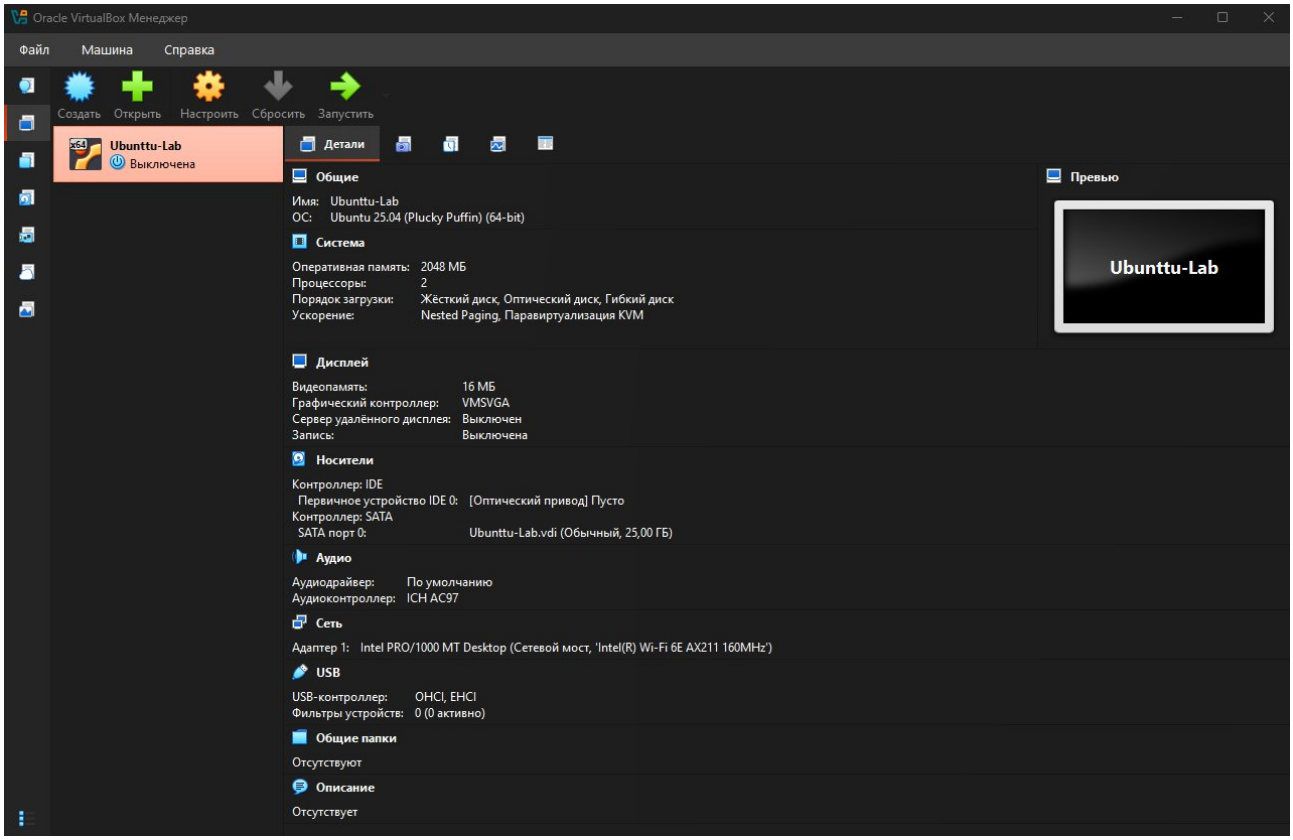
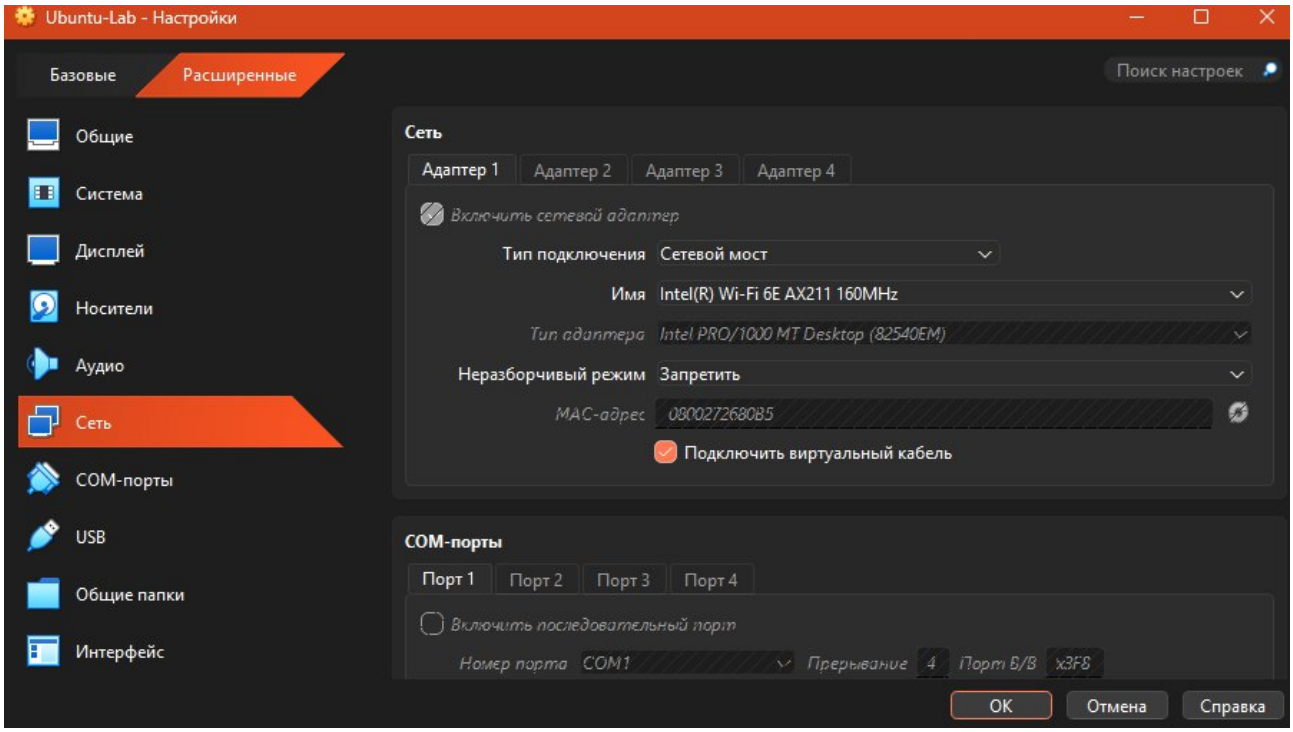
Лабараторная № 4
Тема: Развёртывание машины ОС Linux (Debian/Ubuntu) в Oracle VM VirtualBox, настройка сети в режиме Bridge и подготовка к установке Zabbix

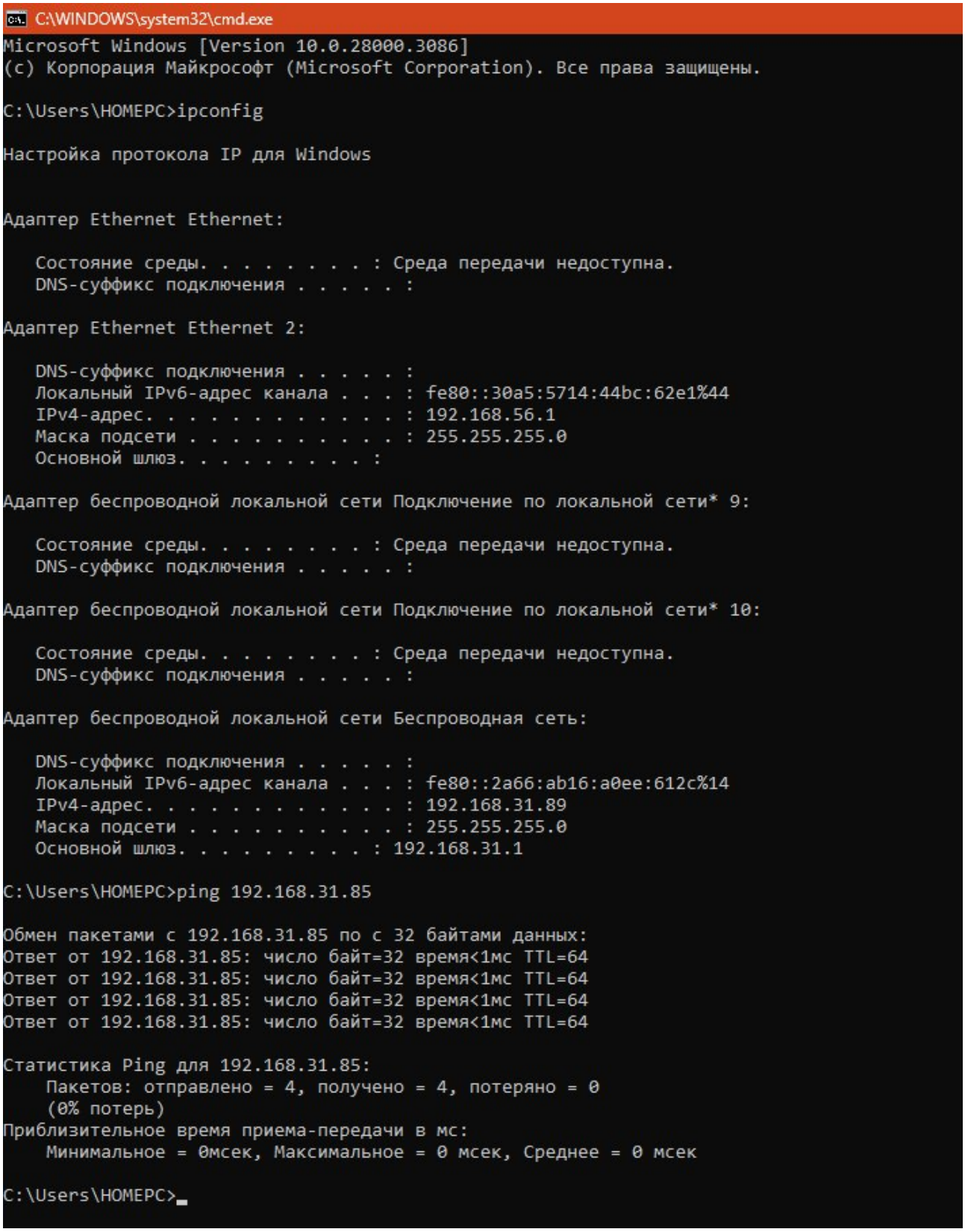
Раздел 1: Создание ВМ в VirtualBox и установка ОС Linux



Раздел 2: Настройка сети в режиме Bridge



Раздел 3: Проверка IP-адресов и ping



```
Ubuntu 26.04.1 LTS Ubuntu-Lab tty1

Ubuntu-Lab login: vboxuser
Password:
Welcome to Ubuntu 26.04.1 LTS (GNU/Linux 7.0.0-42-generic x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:     https://landscape.canonical.com
 * Support:        https://ubuntu.com/pro

 System information as of Thu Nov  5 06:56:24 PM UTC 2026

  System load:  0.0               Processes:              114
  Usage of /:   14.2% of 24.44GB  Users logged in:        0
  Memory usage: 15%               IPv4 address for enp0s3: 192.168.31.85
  Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

45 updates can be applied immediately.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

vboxuser@Ubuntu-Lab:~$ ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000
    link/ether 08:00:27:26:80:b5 brd ff:ff:ff:ff:ff:ff
    altname enx0800272680b5
    inet 192.168.31.85/24 metric 100 brd 192.168.31.255 scope global dynamic enp0s3
       valid_lft 42716sec preferred_lft 42716sec
    inet6 fe80::a00:27ff:fe26:80b5/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
vboxuser@Ubuntu-Lab:~$
```

```
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status



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    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
       valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host noprefixroute
       valid_lft forever preferred_lft forever
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    link/ether 08:00:27:26:80:b5 brd ff:ff:ff:ff:ff:ff
    altname enx0800272680b5
    inet 192.168.31.85/24 metric 100 brd 192.168.31.255 scope global dynamic enp0s3
       valid_lft 42716sec preferred_lft 42716sec
    inet6 fe80::a00:27ff:fe26:80b5/64 scope link proto kernel_ll
       valid_lft forever preferred_lft forever
vboxuser@Ubuntu-Lab:~$ ping -c 4 192.168.31.89
PING 192.168.31.89 (192.168.31.89) 56(84) bytes of data.

--- 192.168.31.89 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3087ms

vboxuser@Ubuntu-Lab:~$ ping -c 4 192.168.31.89
PING 192.168.31.89 (192.168.31.89) 56(84) bytes of data.

--- 192.168.31.89 ping statistics ---
4 packets transmitted, 0 received, 100% packet loss, time 3075ms

vboxuser@Ubuntu-Lab:~$ ping -c 4 192.168.31.89
PING 192.168.31.89 (192.168.31.89) 56(84) bytes of data.
64 bytes from 192.168.31.89: icmp_seq=1 ttl=128 time=0.663 ms
64 bytes from 192.168.31.89: icmp_seq=2 ttl=128 time=0.384 ms
64 bytes from 192.168.31.89: icmp_seq=3 ttl=128 time=0.310 ms
64 bytes from 192.168.31.89: icmp_seq=4 ttl=128 time=0.384 ms

--- 192.168.31.89 ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 3035ms
rtt min/avg/max/mdev = 0.310/0.435/0.663/0.134 ms
vboxuser@Ubuntu-Lab:~$
```

Раздел 4: Ознакомление с инструкцией по установке Zabbix

В ходе выполнения лабораторной работы была успешно развернута виртуальная машина с ОС Ubuntu в среде Oracle VM VirtualBox. Виртуальной машине были выделены необходимые ресурсы (2048 МБ ОЗУ, 2 ядра ЦП). Сетевой интерфейс был настроен в режиме «Сетевой мост», что обеспечило прямую сетевую связность между виртуальной машиной и основным ПК. Двусторонняя проверка связи (ping) подтвердила корректность настройки сети. Также была изучена официальная инструкция по установке Zabbix, что является подготовительным этапом для дальнейшего развертывания системы мониторинга. Цель работы достигнута.